

Poisson Distribution

Description:

The Poisson distribution is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space, provided that these events happen with a known constant mean rate and are independent of the time since the last event.

Formula:

The probability mass function (PMF) of the Poisson distribution is given by:

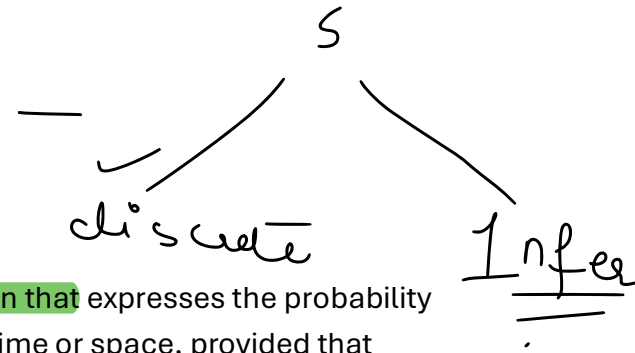
$$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

where:

- $P(X=k)$ = probability of observing k events,
- λ = average number of events in the interval (rate parameter),
- e = Euler's number (approximately 2.71828),
- k = number of events ($k = 0, 1, 2, \dots$).

Applications:

- **Telecommunications:** Modeling the number of phone calls received by a call center in an hour.
- **Traffic Flow:** Counting the number of cars passing through a toll booth in a specified time.
- **Natural Events:** Number of earthquakes in a region over a year.



Exponential Distribution

Description:

The Exponential distribution is a continuous probability distribution that describes the time between events in a Poisson process. It models the time until the next event occurs, assuming that the events occur continuously and independently at a constant average rate.

Formula:

The probability density function (PDF) of the Exponential distribution is given by:

$$f(x; \lambda) = \lambda e^{-\lambda x} \text{ for } x \geq 0 \quad \text{or} \quad f(x; \lambda) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

where:

- $f(x; \lambda)$ = probability density function,
- λ = rate parameter (inverse of the mean),
- x = time until the next event.

Applications:

- **Queueing Theory:** Modeling the time between arrivals of customers at a service point (e.g., a bank or supermarket).
- **Reliability Engineering:** Estimating the time until failure of a mechanical system or component.
- **Survival Analysis:** Analyzing the time until an event occurs, such as death or disease progression in medical studies.